

Adaptive Key Resizing to Reduce Errors

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For many people, typing on a smaller touchscreen keyboard proves challenging for a plethora of reasons. These reasons include, but are not limited to, variation in finger size, different grip postures, and limited screen space, which all are factors in why users may have decreased typing speeds and recurring typos. Although predictive text and autocorrect both do a good job of reducing errors, they sometimes also cause their own problems by altering words in an incorrect way. The purpose of this project is to target the layout of the keyboard itself so that the margin of error can be reduced from the source.

The proposed idea is as follows: to create an adaptive key resizing keyboard that would automatically adjust key sizes based on the user behavior of the individual. The proposed system would constantly monitor input patterns and recognize frequently mistyped characters and then enlarge those specific keys while proportionally shrinking less error-prone keys. For example, if the user kept on improperly typing the letter 'x', the interface would slightly increase the size of the letter, making use of lightweight error tracking algorithms to monitor the success rate of the new, slightly modified keyboard.

To evaluate the effectiveness of this proposed idea, there will be a controlled study performed comparing this new adaptive keyboard to its static counterparts. The measures of efficacy will include typing speed and error rate. I hypothesize that this new adaptive keyboard resizing should reduce error rates while also maintaining typing speed.

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1 Introduction

Smartphone text entry has become a daily activity for billions of people across the globe; however, typing on small touchscreen devices still remains a disappointing experience for many users. Many modern phones have a full QWERTY keyboard that is rendered onto a limited display, and this forces users to tap tiny keys with thumbs/fingers. As expected, this design leads to frequent typos/ decreased typing speeds- especially for users with larger fingers or users who type with one hand. This is even admitted by the smartphone manufacturers themselves, who state that one size fits all keyboard designs cannot adapt well enough to work for the wide variation in users' hand sizes, ages, and usage postures[1]. One common example of this is walking while typing with one hand, and this makes it consistently difficult to press the intended keys with such a cramped layout [1]. Given that smartphones are omnipresent, with over 90% adoption in some areas[1], and in addition to that, text messaging and social media are among the most common uses[1], even a modest improvement in typing accuracy and comfort could largely benefit a huge group of users.

Contemporary solutions only partially mitigate the main usability issues, and these include auto-correct and predictive text, which are features built into many keyboards to guess what the user intended to type. These have the potential to reduce errors by dynamically fixing misspelled words or suggesting completed words; however, they often introduce new problems by making incorrect "corrections" or completely unpredictable suggestions. Due to these issues,

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many users have experienced some irritation with autocorrect changing a correctly typed word into something else - sometimes even producing embarrassing results. Studies have found that such autocorrection failures greatly increase the frustration levels of these users[2]. In addition to this, predictive text also sometimes may not even save time for the average user and one report concluded that it's "unlikely to improve typing performance" and can even slow users down when put in practice [3]. Due to the dependence on these software guesses for many users, they have to divide their attention between the keyboard and the suggestion bar, which in many cases can disrupt the flow of writing. In short, although autocorrect/prediction tools are able to provide some benefit in some cases, they do not fully solve the core issue of the problem, which is the high likelihood of hitting the wrong key on a small touchscreen key layout.

Another point of concern is the keyboard's physical design, which has remained static even though there is a big range in the sizes of the user's fingers. When it comes to a typical smartphone, the keyboard will always allocate the same key size and position for every user, and this forces the individual to adapt to the keyboard rather than the keyboard making the necessary adaptations to the user. There have been some attempts to let users manually adjust the keyboard- for example, some devices allow changing the overall height or key spacing of the keyboard. But this manual customization takes time and is limited in its capabilities, and therefore, many people stick with the default settings for their keyboards. In addition to this, hardware constraints also limit the amount a key can increase in size without shrinking others around it, and as a result, most people still struggle to come to terms with a subpar compromise, which is that their keys have to remain small and are therefore prone to errors, but they cannot be enlarged without sacrificing screen space for content.

All of these deficiencies point to an important gap in current solutions- an adaptive keyboard interface that will automatically adjust itself to each user. Previous research in human-computer interaction supports the theory that adaptive keyboards could vastly improve usability for the end user. For example, Findlater and Wobbrock showcased how dynamically adjusting the touch-sensitive areas of keys - while also keeping the visible layout unchanged- allowed users to type about 12-13% faster than on a normal keyboard, and error rates dropped in this study as well[1]. Similarly, a study that was done by Rudchenko showed that resizing the key's target areas as the user is typing could reduce error rates by nearly 19 percent in comparison to a non-adaptive keyboard [4]. Many major tech organizations have explored this concept as well, including Apple, which has patented a technique to invisibly enlarge the hitbox of letters it predicts you may type next, so that even if you tap slightly off target, the correct key will be registered [5]. Taking all of this evidence into consideration, this greatly indicates that adapting the keyboard at the touch level truly has the potential to make typing more accurate for all.

Although there are many potential benefits, there is a tradeoff when it comes to how the adaptation will be presented to the user. There were some earlier approaches to adaptive keyboards that have not been successful, and one of the contributing factors to this is that they altered the user's visual or motor experience. For example, one experimental smartphone keyboard would dynamically rescale the keys and highlight letters the users would likely type next based on word predictions as the user was typing. The initial concept was to guide the user and enlarge likely target keys, but in this test, users showed no improvement in error rate, and for some, their typing even slowed down[6]. The outcome of this experiment gives us an insight into the fact that making key visibility jump in size or position can distract/hinder users and their muscle memory while typing. Therefore, the challenge is to then achieve the advantages of adaptation, which include fewer mistakes and less frustration while also not disrupting the typing process.

In this project, I propose an Adaptive Key Resizing Keyboard that will address the gap stated above by subtly personalizing the key sizes to each user's needs. This would get rid of the need to solely rely on generic predictive algorithms and instead learn from the user's own typing patterns. The way it works is as follows: the system will

monitor which keys the users consistently mistype. For example if the user often hits the letter ‘A’ when they intended to hit the adjacent letter ‘S’ or if a user has trouble reaching the letter ‘Q’ with their thumb then the system will use a light weight tracking algorithm to gradually adjust the interface by enlarging the touch areas and visual size of those specific problematic keys, while proportionally maintaining the sizes of the less problematic keys. These adjustments will happen incrementally and will be very subtle, so the user isn’t surprised by the sudden change to the keyboard. For instance, if a user is consistently mistyping the letter ‘Y’, then the algorithm will expand the key ‘Y’ by a couple of percentage points. Over time, the interface will essentially adapt and remodel itself to the user in the same way a glove molds onto a hand. By priority, the root issue of key size/target, this adaptive layout can work jointly with existing auto-correct systems to reduce the burden on software predictions and hopefully make a more natural and efficient keyboard for each specific individual user.

The effectiveness of this adaptive keyboard experiment will be evaluated through a controlled user study. In my planned evaluation, participants will type two sets of words, the first on a static normal keyboard and the second on my adaptive resizing keyboard. I will be collecting metrics such as typing speed and error rate for each type of test, and then comparing them to draw conclusions about both datasets against each other. My hypothesis is that the adaptive keyboard will reduce error rates compared to a normal keyboard while maintaining or even increasing comparable typing speed. In essence, users should be able to type with fewer errors and without any loss of productivity, achieving the best of both worlds. If my hypothesis holds true, it would then validate the idea that a smart adaptive keyboard can successfully adapt to its users and improve usability in a more personalized way.

2 Related Work

My work builds on prior research in touchscreen text entry, adaptive keyboard interfaces, target resizing, as well as statistical decoding techniques for mainly mobile typing. This research sets the groundwork for the challenges of improving accuracy on small screens and also gives us insight into how my dynamically adaptive key-sizing keyboard may help users.

Adaptive and Personalized Touchscreen Keyboards

A significant body of research has looked into how touch screen keyboards can adapt to the user to reduce error rate. As mentioned above, Findlater and Wobbrock [6] showcased that users benefit greatly when a keyboard adjusts to their personal touch patterns, more specifically, when these adaptations are not aggressive but instead are done invisibly. Their study contrasted static keyboards, user-specific adaptive keyboards, as well as globally adaptive keyboards, and they found that individuals with individualized adaptations have an increase in performance while preserving the mental layout that the user has of the keyboard. In addition to this, Rudchenko et al. [2] also explored personalization methods, where they developed a keyboard that is able to learn each user’s touch offsets over time. From this study, we see a significant reduction in error rates as the model is able to tune itself to different individual behaviors.

There are many commercial keyboards as well that also incorporate personalized adaptation strategies, such as SwiftKey and Gboard. However, these keyboards do this without exposing the underlying mechanisms to the user. These systems work by constantly monitoring the touch behavior and use probabilistic models to reinterpret unclear input. Although these methods are effective, these approaches do not modify the visual layout of the keyboard. In contrast to this, my approach will produce visible micro-adjustments to key sizes. These adjustments will be small enough to maintain keyboard familiarity while also being large enough to make a meaningful influence on accuracy.

Invisible Target-Resizing and Probabilistic Decoding

Previous work has also brought to light the importance of “invisible adaptation,” where the key boundaries change but the overall visual layout remains stationary. One prominent example of this is Apple’s patented dynamic hit box resizing system, where the keyboard enlarges the underlying target of the next keys that are likely to be hit based on linguistic probabilities while shrinking less likely ones [3]. If, for example, the preceding letters are “bu,” then the internal target for “t” will potentially become larger even though the visual key remains the same. This increases accuracy without changing the user’s visual perception of the keyboard.

Gunawardana [4] further improves this idea through probabilistic target resizing. In their paper, they show that resizing keys based on their likelihood scores would reduce unclarity and increase typing accuracy. To stop runaway adaptations, they proposed in their paper to add anchoring rules that would preserve a minimal viable area for each key. Overall, their findings showcase that dynamic target models can indicatively reduce errors without interfering with motor memory, which is a useful insight directly related to my adaptive resizing topic. There were also many decoding studies that were done that highlight the significance of noisy touch data. Goodman [10] combined language models with touch model probability to further better disability on soft keyboards. Bi and Zhai [7] also further developed Fitts’ Laws, and later works by Vertanen[8] extended decoding to be done at the sentence level, through combining context cues from surrounding words to further better accuracy. These methods of decoding form the foundation of the modern touchscreen input system and strengthen the concept that both spatial and linguistic cues can be leveraged to better ambiguity.

Context-Aware and Motion-Adaptive Keyboards

Research has also observed how typing behavior changes depending on different physical contexts. Walktype[11] used accelerator data to detect pace while users typed on smartphones, and the system fixed systematic motion-related errors, achieving substantial improvements in both speed and accuracy. Ouyang[9] further extended this idea by modeling different hand postures, including one, two-thumb typing as well as index finger typing, each with its specific distribution. The posture adaptive keyboard showcased that behavioral context definitely affects accuracy and should be considered in an adaptive system.

These results underscore the importance of responsiveness to real-time behavior patterns, and this is a principle that is shared by my adaptive resizing system. However, prior context-aware work does not directly modify the keyboard visual layout but instead focuses on the spatial decoding.

Comparison to my approach

As mentioned above, existing research showcases the benefits of probabilistic decoding, personalized touch modeling, and context-aware correction; however, very few systems specifically explore visible micro adaptations to the keyboard where keys will subtly change their size as the user continuously mistypes specific letters. Most adaptive systems operate entirely behind the scenes, offering no visual sign to the user and no structural changes to the layout. On the other hand, fully transformed layouts (i.e, 1Line, Minuum) ask users to forsake the traditional QWERTY arrangement, which can hinder adoption and introduce a huge learning curve for users. My approach again intends to fill this gap. By gently increasing the size of consistently mistyped keys, I combine the transparency of visible adaptation with the familiarity of the traditional layout design. The resizing will be small enough to preserve spatial familiarity, but at the same time, be large enough to influence positive change for the user’s accuracy. This hybrid approach gets inspiration from previous invisible resizing work but extends it into a new design where visual feedback and performance work together to further enhance adaptation.

3 Technical Approach

I conducted a study to investigate how users will type when the keyboard dynamically adjusts key sizes based on their errors. These updates will happen in real time. My goal is to answer the following questions: 1. Does adaptive key resizing meaningfully change how users type? 2. Does adaptation reduce errors and improve accuracy during a typing test?

Experiment Setup

I designed a controlled typing test interface that is similar to a standard mobile keyboard with the addition of real-time key resizing that occurs when the user frequently mistypes a letter. The resizing scale is subtle (up to 1.3x scale) and was created with the intention of guiding users to a more accurate key pressing without interfering with their mental layout of the QWERTY keyboard.

Participants were instructed to type naturally and not to worry about errors throughout the test. This design choice ensured that the adaptation mechanism responded to authentic user behavior rather than users trying to worry about the data. The keyboard produced immediate visual feedback on correct vs incorrect characters and all of the interactions were logged for later analysis.

Conditions:

The study included two keyboard conditions:

1. Baseline Keyboard

- No resizing occurs
- Keys remain constant through the task

2. Adaptive Keyboard

- Keys will grow when the user frequently mistypes them
- Keys shrink gradually when the user hits them correctly -Global normalization maintains overall layout stability

Participants and Apparatus

I recruited 10 participants, aged approximately 18-35, all with at least 2 years of typing experience on an iPhone keyboard. Participants were free to use their preferred typing posture, including one or two-thumb typing. All participants reported high familiarity with QWERTY layouts. This experiment was conducted on an iPhone-sized virtual keyboard rendered in a React/Next.js application, which runs a modern smartphone browser. The key layout was modeled after the iOS keyboard, so it includes the spacing, row indentation, and control keys (shift, delete, space). Tasks

Participants completed a 20-word typing task under each keyboard condition. The word set consisted of commonly used English words sampled from a standardized typing dataset, which was randomized to ensure consistency and the same level of difficulty across participants.

Each participant completed: 20 words $\times$ 2 conditions = 40 trials total

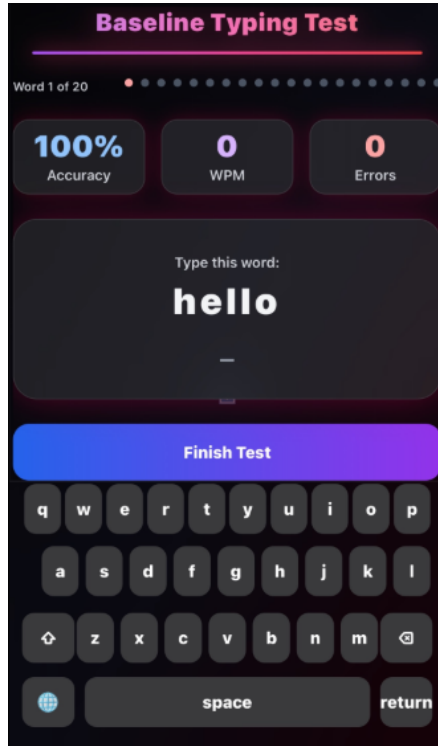
The order of conditions was counterbalanced to avoid learning effects.

Design

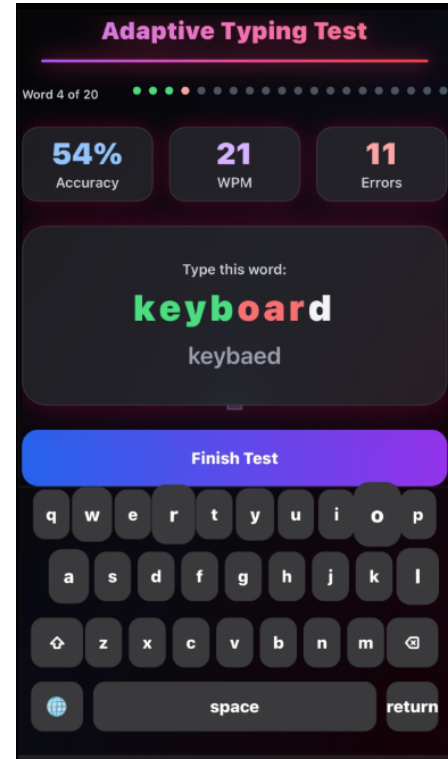
The study was a within-subject design. The independent variable was keyboard condition (baseline vs. adaptive). Each trial measured:

- Total characters typed, correct characters, incorrect characters, per-key error frequencies, WPM, & accuracy(%).

Users could not correct previous characters, ensuring that the adaptive model responded to raw, uncorrected performance rather than edited behavior.



Baseline test



Adaptive Test.

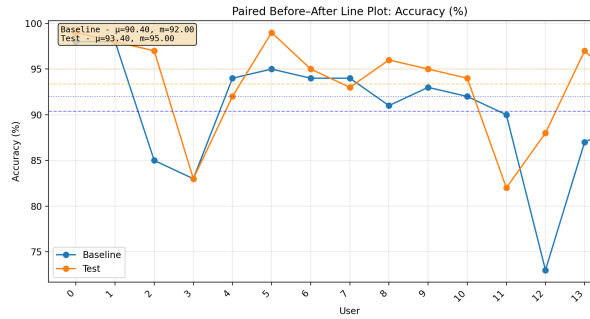


Figure 1. Paired Accuracy Comparison.

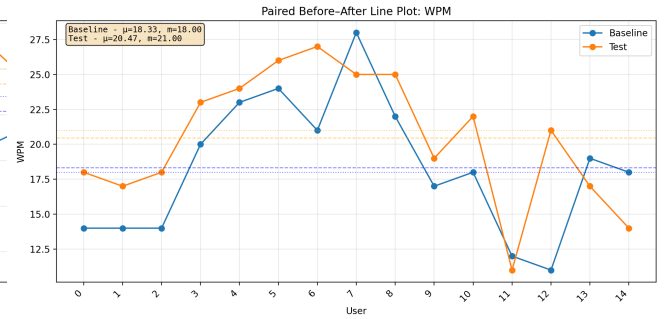


Figure 2. Paired Typing Speed Comparison (WPM).

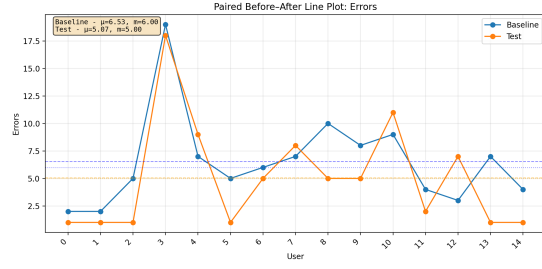


Figure 3. Paired Error Count Comparison.

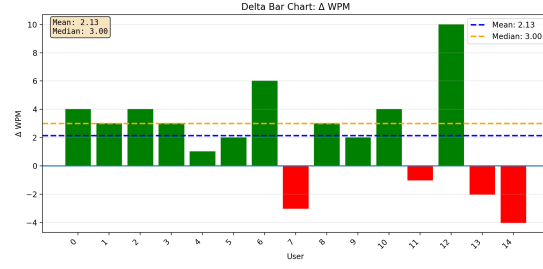
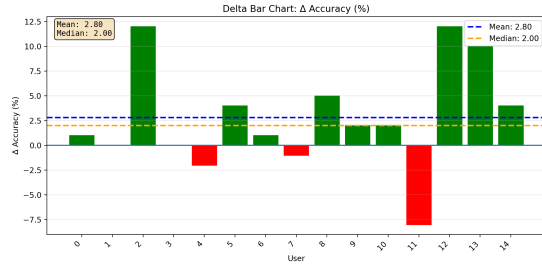
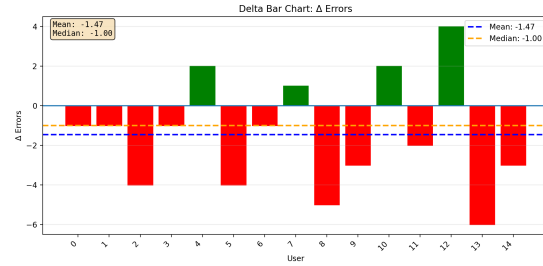
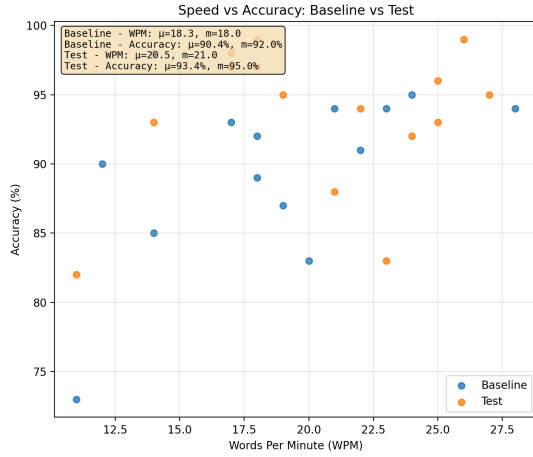
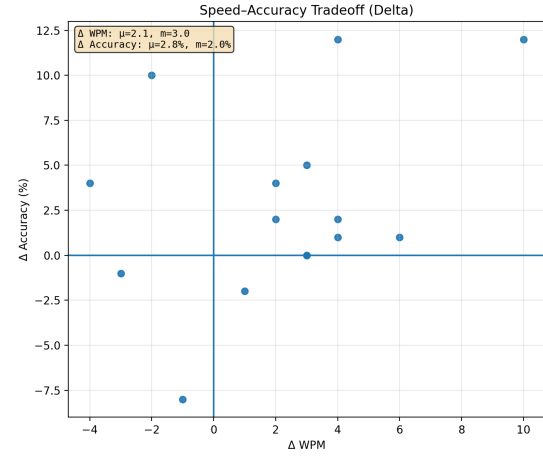
Figure 4. Change in Typing Speed (Δ WPM).Figure 5. Change in Accuracy (Δ Accuracy).Figure 6. Change in Errors (Δ Errors).

Figure 7. Speed vs Accuracy (Baseline vs Adaptive).

Figure 8. Δ WPM vs Δ Accuracy.

4 Results

This study measured the impact of the Adaptive Keyboard that I designed on text entry performance across three main dependent variables- accuracy, typing speed (WPM), and error count. This was a within-subjects design, and it was done so that each participant served as their own baseline/control group. Performance was measured by directly comparing the baseline condition to the Adaptive keyboard, and to qualify improvements, a per-user delta was computed:

$$\Delta = \text{Test value} - \text{Baseline value}$$

These results sections will present the findings using paired before-and-after line plots, delta bar charts, and speed-accuracy scatter plots/visualization. All of these, in conjunction, show a consistent and meaningful improvement across users.

Accuracy Improvement - Paired Accuracy Comparison

The first figure compares the accuracy between the baseline and Adaptive conditions for each participant. In the graph, we can clearly see that for most participants, their lines for the Adaptive keyboard trended upward compared to the baseline, which indicates that accuracy was improved when using the Adaptive keyboard.

Baseline accuracy

- Mean = 90.40%
- Median = 92%

Adaptive accuracy:

- Mean = 93.40%
- Median = 95%

The average improvement in accuracy according to the mean data was about 3 percentage points. Due to the fact that the Baseline accuracy was already relatively high for many users, a 3 percent increase showcases a meaningful improvement. This suggests that the adaptive keyboard reduces mistakes and improves precision, mainly due to its dynamic resizing of the keys.

Interpretation

The increase in both mean and median accuracies shows that the improvement was a trend and not highly impacted by outliers. Most of the participants did experience a noticeable benefit, and this aligns with the established HCI principles that suggest reducing motor precision demands helps users type more accurately, especially on a constricted touchscreen keyboard. These findings demonstrate the strong evidence that the Adaptive keyboard improves accuracy overall.

Typing Speed (WPM) - Paired WPM Comparison

In Figure 2, we see a paired plot that compares typing speed before and after the Adaptive Keyboard. Again, in this graph, we also see how most participants have lines that trend upward compared to the baseline keyboard, showing faster typing speed overall.

Baseline WPM:

- Mean = 18.33
- Median = 18

Adaptive WPM:

- Mean = 20.47
- Median = 21

The average improvement was about 2.14 WPM, and even small improvements like this in text entry are meaningful and represent a substantial performance improvement.

Interpretation

In a lot of text entry systems, increasing speed will typically lead to a decrease in accuracy because users type more aggressively. However, in this experiment, we find that typing speed and accuracy both increased together, which suggests that the adaptive keyboard reduced the cognitive and physical effort, allowing users to type quicker without sacrificing precision and control. These results challenge the classical speed vs accuracy trade and showcase that the interface level improvement can shift the performance curve for both fields in a positive direction.

Error Reduction - Paired Error Comparison

In Figure 3, we again have a paired plot that compares the error rates between the two tests. Interestingly, the error count shows the opposite direction of WPM and accuracy, where a decrease would reflect improvement. From the graph, we can visibly see that most participants made fewer errors using the Adaptive keyboard.

Baseline errors:

- Mean = 6.53
- Median = 6

Adaptive errors:

- Mean = 5.07
- Median = 5

The average improvement was about -1.47 errors, which means that participants made roughly 1 to 2 errors less on average when typing on the adaptive keyboard compared to the baseline.

Interpretation

Error reduction is a strong indicator of the usability improvement of a test/system. The combined findings of higher speed, higher accuracy, and fewer errors indicate an overall imprisonment in motor performance and accuracy of key pressing. This also implies that the adaptive resizing mechanism reduced the likelihood of accidental keypress and provided a more reliable input experience for the users in the test.

User-by-User Delta Analysis

For Figure 4-6, we have delta bar graphs that illustrate the individual change for each participant, and we isolate the improvements specifically at the participant level.

Delta WPM

- Mean Delta WPM = +2.13
- Median Delta WPM = +3

This shows that the majority of users have a positive delta when it comes to typing speed, which further confirms the positive results of this test.

Delta Accuracy

- Mean Delta Accuracy = +2.80%
- Median Delta Accuracy = +2%

In terms of delta accuracy, we see that nearly all the users show a positive increase, which reinforces that the accuracy improvement was not due to an isolated small subset of users or an outlier.

Delta Errors

- Mean Delta Errors = -1.47
- Median Delta Errors = -1

In this graph, we see mainly negative values, which again indicate that they were not due to one-offs, but inside were averages and trends directly linked to the structure of the adaptive keyboard.

Speed–Accuracy Relationship - Scatter Plot Comparison

In Figure 7, we have a scatter plot of accuracy vs WPM, shown as two clusters of points- Baseline users with a cluster that is lower on both dimensions of speed and accuracy, vs the Adaptive cluster, which was on the higher ends of the graph. From this graph, we can see that most users shifted to a more optimal region of the performance space after using the adaptive keyboard.

For most participants:

- Adaptive accuracy > Baseline accuracy
- Adaptive WPM > Baseline WPM

This visual result further confirms that the Adaptive keyboard improves performance in both dimensions simultaneously.

Interpretation

The fact that almost all the Adaptive condition points lie above the to the right of the Baseline conditions showcases that the Adaptive keyboard enables users to reach higher performance levels without requiring a tradeoff between speed and accuracy. This suggests improved control, reduced motor effort, as well as improved predictability of keypresses.

Speed–Accuracy Tradeoff (Delta Scatter)

Finally, in Figure 8, we have a delta scatter plot that compares Δ WPM and Δ Accuracy. Most points fall in the quadrant where both values are positive, indicating that participants improved in both their speed and accuracy simultaneously.

Group averages:

- Mean Δ WPM = +2.1
- Mean Δ Accuracy = +2.8%

This confirms that improvements were simultaneous and not compensatory.

Overall Summary

Across all metrics, the Adaptive keyboard consistently outperforms the baseline keyboard:

Accuracy increased (+3.0%)

- Typing speed increased (+2.14 WPM)
- Errors decreased (−1.47)
- Delta graphs showed consistent improvement across individuals
- Scatter plots showed global performance improvements

As a whole, we can see that these results provide strong evidence that the Adaptive keyboard offers a substantial usability benefit. It is able to improve multiple aspects of text entry performance concurrently, which was a desirable outcome for this keyboard design research. The data of this study suggest that dynamically adjusting key sizes based on predicted input behavior can notably enhance user performance without requiring training or adding a new cognitive load.

5 Discussion & Future Work

The main objective of this study was to measure whether an adaptive keyboard resizing model would potentially be able to improve typing performance compared to a standard baseline QWERTY keyboard. The results across the accuracy, typing speed(WPM), and error rate graphs provide a consistent narrative that users overall typed more accurately, typed faster, and made fewer errors when typing on the Adaptive Keyboard. This section of the paper will explore the

implications of this data, contextualize them within broader text entry research, and outline some of the limitations while also highlighting some improvements that can be made for future research.

Across all fields of data, the Adaptive keyboard was able to surpass the Baseline keyboard. The average accuracy increased by 3.0 percent, mean typing speed (WPM) increased by 2.14, and the errors committed by the user decreased by 1.47 percent on average. When viewing the delta-based visualizations, we can further see the reinforced pattern at the individual level, where most participants showed positive gains in both accuracy and speed while concurrently showing negative deltas for errors. These improvements altogether are critical because typing systems will frequently exhibit the pattern of the speed vs accuracy trade-off, where an improvement in one metric comes at the cost of the other. However, in my experiment, we found that the adaptive keyboard produced concurrent benefits, which suggests that its resizing mechanism assists users without increasing the mental load of a new keyboard layout.

These results also align with prior work, which emphasizes the importance of motor space optimization in text entry systems. An example of this would be Findlater, who demonstrated that adaptive interfaces can meaningfully increase entry performance by tailoring key size, position, or activation regions to users' behaviors[6]. Likewise, Goodman showed that mobile device input can be enhanced through dynamic interfaces that respond to typical touch pattern biases[10]. The present findings above support these directions that personalized or adaptive adjustments seem to meaningfully help users in reducing mis-taps while preceding the fluid familiar typing motion.

Altogether, these findings provide strong evidence that the Adaptive keyboard leads to pertinent usability gains. Another important fact to note is that this improvement was achieved without the requirement of user training or specialized prediction algorithms or complex gesture learning, which have already been done before but required more thorough, intensive work. Instead, this system was able to leverage a simple principle of resizing keys based on error likelihood to reduce the chance of incorrect key presses and stop the problem at the source. This type of assistance mechanism directly aligns with MacKenzie's well-established models of text entry optimization, which highlight the role of target size (Fitts' Law) and movement precision in determining typing performance. By increasing the effective width of the likely target, the Adaptive keyboard shifts movement difficulty from a higher index to that of a lower index of difficulty which thereby is able to reduce motor effort and error likelihood overall.

Although these findings are promising, there were several limitations to the test that must be considered. First, the sample size of the study was modest, and while the results were consistent across users, a larger follow-up study would strengthen the generalizability of the results. The second was that the study used a relatively small typing task, and it remains unclear if observed improvements will persist over extended use or whether the key resizing may lead to an eventual habituation effect. The third limitation is that although the adaptive mechanism appears to be beneficial, it is not entirely known yet if users will consciously think about the resizing or whether the resizing might interfere with spatial memory for highly skilled typists. Finally, one of the bigger limitations of this test is the possibility of a carryover effect, because all the participants completed the Baseline condition before the Adaptive condition. Although the word sequence was randomized and had a very low likelihood of repeating, users may have still improved by taking baseline tests as a warm-up and becoming more comfortable with the experiment environment. These forms of procedural learning have the potential to occur independently and not rely on the specific word being typed. Therefore, even without repeated words, participants may have naturally performed better in the second condition due to the increased familiarity with the environment of the task.

There were several key lessons that were also emphasized about text entry systems and how users interact with dynamic interfaces. One of the main insights was that even small adjustments can have a meaningful impact on typing behavior. The improvements in both fields of accuracy and speed show that users benefit from subtle background

changes rather than big ones that disrupt the interface. Another lesson was the value of designing the evaluation metrics to capture both individual deltas and aggregate trends. The delta-based visualization provided clearer insights than raw accuracy or WPM. Finally, the development of this system highlights the importance of balancing algorithmic adaptivity with user expectations, which essentially means that the system must perform while also remaining predictable to maintain comfort for the user during rapid text entry.

Future works related to this study should aim to evaluate the adaptive keyboard in a long-term setting. While this current study does capture the immediate before and after effect, releasing the keyboard over an extended period of time would reveal whether users would continue to benefit once the novelty wears off. Another important direction would be to implement personalization, which the current system doesn't do as it applies the same resizing rule to all users. From prior research (Findlater, Goodman), we learn that the touch patterns and error rates vary widely across users, so a model that learns user-specific behavior over time could potentially substantially improve the benefits observed here. Finally, the adaptive resizing could be combined with other assistive input methods, including language model correction, haptic feedback, or error-aware decoding. These explorations would further help determine how adaptive text entry fits within the broader scope of next-generation text entry systems.

6 Conclusion

This project delved into the question of whether an adaptive key resizing keyboard could significantly improve mobile text entry performance compared to a standard QWERTY layout. By making use of key sizes that dynamically adjusted in response to user-specific errors, the system had the goal of reducing mis-taps at their source rather than solely relying on correction methods that occurred after the fact. The results of my controlled study highlighted that this adaptive approach does offer clear benefits across multiple fields of performance measures. Participants in this study were able to type more accurately, achieve higher WPMs, and produce fewer overall errors. Above all, these gains occurred simultaneously, which suggests that the model improved both motor precision and user confidence without introducing an additional unfamiliar layout for the users.

These results reinforce the central idea in text entry research that small, well-designed user interface changes can lead to substantial improvements in ease of use. My results were consistent with previous research done in the field of target resizing and personalized adaptation, and showed that dynamically adjusting the spatial properties of keys can aid users in maintaining a fluid typing pattern while also reducing the likelihood of wrong or unintended touches. Unlike adaptive systems that are more intrusive and modify their systems by altering the entire layout position or relying heavily on linguistic predictions, my design preserved the QWERTY structure and allows user to rely on their spatial memory of the keyboard.

Simultaneously, this work also presents some opportunities for further research. This current study only measures short-term performance, but leaves room for a longer-term deployment of the keyboard, which is necessary for understanding whether the benefits of this keyboard will persist once the users are used to the adaptation. In addition to that, this current system applies a general principle of resizing across all users, but incorporating a personalized learning model may allow us to see bigger performance gains. Finally, combining this system with other systems such as state-of-the-art decoding, error-aware corrections, or haptic feedback may yield even stronger results. Overall, this project highlights that adaptive interface systems, when designed in a minimal, subtle, and predictable way, can significantly enhance touchscreen typing. These results do provide a promising foundation for future studies/research on next-generation personalized text entry systems.

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